



Nickel and Cobalt Alloys
supplied in Sheet, Plate and Bar

Custom 455® Technical Data

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Type Analysis

Element	Min	Max
Carbon	--	0.05
Manganese	--	2.00
Silicon	--	0.50
Phosphorus	--	0.040
Sulfur	--	0.030
Chromium	11.00	12.50
Nickel	7.50	9.50
Molybdenum	--	0.50
Copper	1.50	2.50
Columbium+Tantalum	0.10	0.50

Description

Recognizing the need for high-strength alloys with good corrosion resistance to atmospheric environments, and a martensitic age-hardenable stainless steel. This alloy is relatively soft and formable in the annealed condition. A single-step aging treatment develops exception-alloy high yield strength with good ductility and toughness.

This stainless can be machined in the annealed condition , and welded in much the same manner as other precipitation hardenable stainless steels. Because of its low work-hardening rate, it can be extensively cold formed. The dimensional change during hardening is only about -0.001 in/in, which permits close-tolerance finish machining in the annealed state.

Custom 455 stainless represents a significant advancement in the area of precipitation hardening stainless steels. It should be considered where simplicity of heat treatment, ease of fabrication, high strength and corrosion resistance are required in combination.

Selection

Because of the unique combination of high strength and corrosion resistance of Custom 455 stainless there are few other alloys available for consideration. PH13-8 Mo* can be considered where good transverse toughness and ductility are necessary in large sections. *PH13-8 Mo is registered trademark of ARMCO, Inc.

Corrosion Resistance

Custom 455 stainless resists staining in normal air atmospheres, and shows no corrosion in fresh water. Test in 5% salt spray at 95 °F(35 °C) and in 5% ferric chloride at room temperature have demonstrated good resistance to pitting and rusting. Laboratory test s in a variety of mild chemical environments have shown that the level of general corrosion resistance is superior to that of the 12% chromium steels (Type 430). In most tests there was no significant effect of aging temperature on corrosion resistance. Hydrogen embrittlement tests in 5% acetic acid saturated with sulfuric hydrogen at room temperature show the same degree of susceptibility as other high-strength martensitic stainless steels. All high-strength steels are subject to stress corrosion under certain stress conditions and in certain environments. Numerous severe tests have been conducted to evaluate the behavior of Custom 455 stainless in different environments, and to determine the effect of aging temperature on resistance to stress -corrosion cracking. These tests include U-bend specimens, direct tension specimens, C-rings and precracked cantilever beam specimens. Environments have included marine atmospheres, 20% salt spray, and 3-1/2% sodium chloride solutions. All tests have shown that Custom 455 stainless has inherently good resistance to stress-corrosion cracking, and that this resistance improves markedly as aging temperature is increased. For optimum corrosion resistance, surfaces must be free of scale and foreign particles and finished parts should be passivated.

Elevated Temperature Use

Custom 455 stainless has displayed excellent resistance to oxidation up to approximately 1100 °F. Long-term exposure to temperatures between about 600-900 °F can result in reduced toughness in precipitaiton hardenable stainless steels. The reduction in toughness can be minimized in some cases by using higher aging temperatures. Short exposures to elevated temperatures can be considered, provided the maximum temperature is at least 50 °F less than the aging temperature.

Physical Properties

(Condition H 950)

Specific gravity..... 7.76

Density

lb/cubic in 0.280

kg/cubic m 7760

Modulus of Elasticity (E)

ksi..... 29.0 x 10(3)

MPa.....200 x 10(3)

Modulus of Rigidity (G)

ksi 11.0 x 10(3)

Mpa x 10(3)..... 75.8 x 10(3)

Poisson's Ration.....0.30

Mean Coefficient fo thermal expansion

Temperature		10(-6)/°F	10(-6)/K
72 °F to	22 °C to		
200	93	5.90	10.6
300	149	6.03	10.9
500	260	6.20	11.20
700	371	6.45	11.6
900	482	6.68	12.0

Thermal Conductivity

Test Temperature		Btu-in/ft²-h-°F	W/m-K
°F	°C		
212	100	125	18.0
392	200	137	19.7
572	300	148	21.3
752	400	162	23.4
932	500	172	24.7

Electrical Resistivity

Condition	Ohm-cir-mil/ft	Microhm/mm
A	545	906
H 950	456	758

Heat Treatment

Condition A

(Solution Treated or Annealed): Heat to 1500/1550 °F, and cool rapidly. Water quenching is preferred for small sections. Custom 455 stainless will normally be supplied from the mill in the annealed condition, ready for the one-step hardening treatment.

Condition H 900, H 950, H 1000, H 1050

(Precipitation or Age Hardened): The high strength levels of Custom 455 are derived from a single precipitation-hardening treatment consisting of heating to a selected temperature between 900/1050 °F, holding for 4 hours and air cooling

Average Longitudinal Size Change (Contracting)

Solution-treated to aged condition

Condition	Contraction in./in.
H 900	0.0007
H 950	0.0009
H 1000	0.0012

Cleaning

Descaling following forging and annealing can be accomplished by acid cleaning or grit blasting. The acic treatment consists of 2 minutes in 50% by volume muriatic acid at 180 °F, followed by 4 minutes in a mixture of 15% by volume nitric acid, plus 3% by volume hydrofluoric acid at room temperature. Repeat cleaning procedure as neccessary but decrease the times by 50% (i.e., 1 and 2 minutes, respectively). The heat tint from aging can be removed by polishing, vapor blasting or pickling 4 to 6 minutes in a mixture of 15% by volume nitric acid, followed by a water rinse. Repeat the acid cleaning procedure if necessary, but decrease the time by 2 to 3 minutes. Desmut in 20% by volume nitric acid at room temperature. After acid cleaning, bake 1 to 3 hours at 300/350 °F to remover hydrogen.

Workability

Hot working

Custome 455 stainless is easily forged within the temperature range fo 1650/2300 °F. For optimum mechanical properties, material to be forged should be heated uniformly to 1900/2100 °F and soaked at heat; finishing temperature should be within the range of 1500/1700 °F to obtain an optimum grain size and properties after heat treating. Cool forging in air to rom temperature and anneal.

Cold working

Cold working is readily performed despite the fact that Custom 455 stainless is martensitic in the annealed condition, and has a hardness of Rockwell C 30/35. Deep drawing or sketching operations will required intermediate anneals because the elongation tends to be localized. For other cold-forming operations such as cold drawing and cold rolling, the work-hardening rate in the annealed condition is extremely low allowing considearble cold working without intermediate annealing. Cold-heading and warm-heading operations are also easily performed. Cold working prior to aging results in even higher tensile and yield strengths in the hardened condition.

Spring Properties

Wire in the annealed or lightly drawn condition and annealed strip can be aged for 2 to 4 hours at 900°F (482°C) to achieve a tensile strength in excess of 250 ksi (1742 MPa). As such, Custom 455 stainless can be used in relatively large diameters and thicknesses and still have good spring characteristics. This greatly expands the range of sizes of corrosion-resistant springs available to designers. Custom 455 stainless can also be extensively cold worked prior to aging to achieve even higher levels of strength. Extensive cold working reduces the aging time required to reach maximum mechanical properties. Therefore, cold-drawn spring wire and cold-rolled strip in Condition C need only be aged for 1/2 hour at 850°F (454°C).

Diameter		Nominal Cold-Drawn Tensile Strength		Age-Hardened Tensile Strength			
inches	mm	ksi	MPa	MIN	MAX	MIN	MAX

Machining

Custom 455 stainless have been machined successfully using the same practices required for other high-strength alloys; i.e., rigid tool and work supports, slower speeds, positive cuts and adequate amounts of coolant. The machining characteristics of Custom 455 stainless are similar to those of the nickel maraging steels. When using carbide tools, surface speed feet/minute can be increased between 2 to 3 times over the high speed suggestions. Feeds can be increased between 50 and 100%.

High Speed Tool				
Turning- Cut-Off And Forming	Cut-Off Tool Width	1/16"	SFPM IPR	60 .001
		1/8"	SFPM IPR	60 .0015
		1/4"	SFPM IPR	60 .002
		1/2"	SFPM IPR	60 .0015
	Form Tool Width	1"	SFPM IPR	60 .001
		1-1/2"	SFPM IPR	60 .0007
Drilling	Drill Dia.	1/4"	SFPM IPR	50 .004
		3/4"	SFPM IPR	50 .008
Reaming	Under 1/2"		SFPM IPR	60 .003
	Over 1/2"		SFPM IPR	60 .008
Die Threading	T.P.I	3-7½	SFPM	5-12
		8-15	SFPM	8-15
		Over 16	SFPM	10-20

Welding

Custom 455 stainless can be satisfactorily welded by the shielded fusion and resistance welding processes. Oxyacetylene welding is not recommended, since carbon pickup in the weld may occur. Preheating is not required to prevent cracking during the welding of this alloy. Normally, welding in the solution-annealed condition is satisfactory; however, where high welding stresses are anticipated, it may be advantageous to weld in the overaged (aged at 1150 °F) condition. If welded in the solution-annealed condition, the alloy can be directly aged to the desired strength level after welding. However, the optimum combination of strength, ductility and corrosion resistance is obtained by solution annealing the welded part before aging. If welded in the overaged condition, the part should be solution annealed before aging.

Typical Mechanical Properties

Typical Room Temperature Mechanical Properties

Bar, 1"(25.4mm) section

Condition	0.2% Yield Strength		Ultimate Tensile Strength		Notch Tensile Strength K(t)=10		% Elong. in 4D	% Reduction of Area	Rockwell C Hardness	Charpy V-Notch Impact Strength	
	ksi	MPa	ksi	MPa	ksi	MPa				ft-lb	J
A	115	793	140	965	--	--	12	50	31	--	--
H 900	220	1516	230	1585	250	1723	10	45	48	8	11
H 1000	195	1344	205	1413	250	1723	12	45	45	12	16
H 1050	175	1206	190	1309	250	1723	14	50	40	25	34

Typical Elevated Temperature

Tensile properties, condition H950

Test Temperature		0.2% Yield Strength		Ultimate Tensile Strength		% Elongation in 4D	% Reduction of Area	Room Temp. Rockwell C Hardness after test
°F	°C	ksi	MPa	ksi	MPa			
Room		220	1517	230	1586	12	50	48
600	316	185	1276	195	1345	12	50	48
800	427	163	1124	175	1207	14	60	48
1000	538	110	758	130	896	18	70	48

Typical Elevated Temperature

Tensile properties, condition H1000

Test Temperature		0.2% Yield Strength		Ultimate Tensile Strength		% Elongation in 4D	% Reduction of Area	Room Temp. Rockwell C Hardness after test
°F	°C	ksi	MPa	ksi	MPa			
Room		195	1345	205	1413	14	55	44
600	316	165	1138	174	1200	14	60	44
800	427	163	1020	154	1062	15	65	44
1000	538	110	758	118	814	20	75	44

Typical Stress Rupture Strength

Test Temperature		Condition	Stress for rupture in					
°F	°C		ksi	MPa	ksi	MPa	ksi	MPa
800	427	H 950	142	979	117	807	91	627
900	482	H 950	109	752	82	565	54	372

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Type Analysis

Element	Min	Max
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Sulfur	--	0.030
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Physical Properties

(Condition H 950)

Specific gravity..... 7.76

Density

lb/cubic in 0.280

kg/cubic m 7760

Modulus of Elasticity (E)

ksi..... 29.0 x 10(3)

MPa.....200 x 10(3)

Modulus of Rigidity (G)

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Mpa x 10(3)..... 75.8 x 10(3)

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Thermal Conductivity

Test Temperature		Btu-in/ft²-h-°F	W/m-K
°F	°C		
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392	200	137	19.7
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Condition	Ohm-cir-mil/ft	Microhm/mm
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Spring Properties

Wire in the annealed or lightly drawn condition and annealed strip can be aged for 2 to 4 hours at 900°F (482°C) to achieve a tensile strength in excess of 250 ksi (1742 MPa). As such, Custom 455 stainless can be used in relatively large diameters and thicknesses and still have good spring characteristics. This greatly expands the range of sizes of corrosion-resistant springs available to designers. Custom 455 stainless can also be extensively cold worked prior to aging to achieve even higher levels of strength. Extensive cold working reduces the aging time required to reach maximum mechanical properties. Therefore, cold-drawn spring wire and cold-rolled strip in Condition C need only be aged for 1/2 hour at 850°F (454°C).

Diameter		Nominal Cold-Drawn Tensile Strength		Age-Hardened Tensile Strength				
inches	mm	ksi	MPa		MIN	MAX	MIN	MAX

Machining

Custom 455 stainless have been machined successfully using the same practices required for other high-strength alloys; i.e., rigid tool and work supports, slower speeds, positive cuts and adequate amounts of coolant. The machining characteristics of Custom 455 stainless are similar to those of the nickel maraging steels. When using carbide tools , surface speed feet/minute can be increased between 2 to 3 times over the high speed suggestions. Feeds can be increased between 50 and 100%.

High Speed Tool

Turning- Cut-Off And Forming	Cut-Off Tool Width	1/16"	SFPM IPR	60 .001
		1/8"	SFPM IPR	60 .0015
		1/4"	SFPM IPR	60 .002
		1/2"	SFPM IPR	60 .0015
	Form Tool Width	1"	SFPM IPR	60 .001
		1-1/2"	SFPM IPR	60 .0007
Drilling	Drill Dia.	1/4"	SFPM IPR	50 .004
		3/4"	SFPM IPR	50 .008
Reaming	Under 1/2"		SFPM IPR	60 .003
	Over 1/2"		SFPM IPR	60 .008
Die Threading	T.P.I	3-7½	SFPM	5-12
		8-15	SFPM	8-15
		Over 16	SFPM	10-20

Welding

Custom 455 stainless can be satisfactorily welded by the shielded fusion and resistance welding processes. Oxyacetylene welding is not recommended, since carbon pickup in the weld may occur. Preheating is not required to prevent cracking during the welding of this alloy. Normally, welding in the solution-annealed condition is satisfactory; however, where high welding stresses are anticipated, it may be advantageous to weld in the overaged (aged at 1150 °F) condition. If welded in the solution-annealed condition, the alloy can be directly aged to the desired strength level after welding. However, the optimum combination of strength, ductility and corrosion resistance is obtained by solution annealing the welded part before aging. If welded in the overaged condition, the part should be solution annealed before aging.

Typical Mechanical Properties

Typical Room Temperature Mechanical Properties

Bar, 1"(25.4mm) section

Condition	0.2% Yield Strength		Ultimate Tensile Strength		Notch Tensile Strength K(t)=10		% Elong. in 4D	% Reduction of Area	Rockwell C Hardness	Charpy V-Notch Impact Strength	
	ksi	MPa	ksi	MPa	ksi	MPa				ft-lb	J
A	115	793	140	965	--	--	12	50	31	--	--
H 900	220	1516	230	1585	250	1723	10	45	48	8	11
H 1000	195	1344	205	1413	250	1723	12	45	45	12	16
H 1050	175	1206	190	1309	250	1723	14	50	40	25	34

Typical Elevated Temperature

Tensile properties, condition H950

Test Temperature		0.2% Yield Strength		Ultimate Tensile Strength		% Elongation in 4D	% Reduction of Area	Room Temp. Rockwell C Hardness after test
°F	°C	ksi	MPa	ksi	MPa			
Room		220	1517	230	1586	12	50	48
600	316	185	1276	195	1345	12	50	48
800	427	163	1124	175	1207	14	60	48
1000	538	110	758	130	896	18	70	48

Typical Elevated Temperature

Tensile properties, condition H1000

Test Temperature		0.2% Yield Strength		Ultimate Tensile Strength		% Elongation in 4D	% Reduction of Area	Room Temp. Rockwell C Hardness after test
°F	°C	ksi	MPa	ksi	MPa			
Room		195	1345	205	1413	14	55	44
600	316	165	1138	174	1200	14	60	44
800	427	163	1020	154	1062	15	65	44
1000	538	110	758	118	814	20	75	44

Typical Stress Rupture Strength

Test Temperature		Condition	Stress for rupture in					
°F	°C		ksi	MPa	ksi	MPa	ksi	MPa
800	427	H 950	142	979	117	807	91	627
900	482	H 950	109	752	82	565	54	372

Custom 455 - Current Inventory Stock

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